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Saveetha Institute of Medical And Technical Sciences
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INTELLIGENT RESEARCH PAPER ANALYZER AND LITERATURE REVIEW ASSISTANT USING NLP AND LARGE LANGUAGE MODELS

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements for
the Course of*

DSA0305 – NATURAL LANGUAGE PROCESSING

to the award of the degree of
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IN

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "Intelligent Research Paper Analyzer and Literature Review Assistant using NLP and Large Language Models" has been carried out by **Narappagari Sravya (Reg. No. 192472192)** as an individual contribution towards the team project undertaken with Sahasra Medarametla (192425317), Sravanthi Duggirala (192425315), N Adi Narayana (192525346) and T Lakshmi Prasad Naidu (192324187), under the supervision of [Guide Name], and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech Computer Science and Engineering programme at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, **Narappagari Sravya(192472192)** of the Department of Computer Science and Engineering, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that this Capstone Project Report entitled "Intelligent Research Paper Analyzer and Literature Review Assistant using NLP and Large Language Models" presents my own bonafide individual contribution towards the team project, specifically the design, development and evaluation of **Module 1 — Secure Upload & Validation**. To the best of my knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity.

All sources, references, data, and other materials used have been appropriately acknowledged. I further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. I confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of this work. This report is submitted individually and is not identical to the reports submitted by the other members of my team.

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INDIVIDUAL CONTRIBUTION STATEMENT

(Mandatory for all team projects)

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ABSTRACT

The exponential growth of scientific publishing has made it increasingly difficult for students and researchers to efficiently read, comprehend, and synthesise large volumes of research papers when conducting a literature review. Manually extracting key findings, methodologies, and research gaps from dozens of papers is time-consuming and prone to inconsistency. This capstone project addresses this engineering problem by designing and implementing an Intelligent Research Paper Analyzer and Literature Review Assistant that uses Natural Language Processing (NLP) and Large Language Models (LLMs) to automate paper comprehension and comparison.

The system is organised into five interacting modules: (1) secure research-paper upload and validation, (2) intelligent PDF content extraction, (3) AI-powered research paper analysis, (4) automated report generation and PDF export, and (5) interactive question answering over paper content. This report documents the individual contribution of the author to Module 3, the AI-Powered Research Paper Analysis engine, which converts extracted paper text into semantically indexed chunks, retrieves the most relevant context for a given analysis task, and uses prompt-engineered calls to a large language model to generate structured summaries, key-finding tags, methodology classification, and research-gap assessments.

The methodology combined literature and product review of existing summarization and literature-review-assistant tools, iterative prompt design, retrieval-augmented generation (RAG) techniques, and quantitative evaluation using ROUGE-L for summary quality, F1-score for key-finding extraction, and precision for gap detection, benchmarked against a TF-IDF/extractive-summarization baseline. On a held-out set of research papers from the Computer Science and NLP domains, the proposed LLM-based module achieved a ROUGE-L score of 76%, key-finding extraction F1 of 82%, gap-detection precision of 74%, and downstream question-answering accuracy of 85%, outperforming the extractive baseline across all measured metrics.

Keywords: Natural Language Processing; Large Language Models; Retrieval-Augmented Generation; Automated Literature Review; Text Summarization; Research Paper Analysis

TABLE OF CONTENTS

Sl. No	Title	Page No
i	CERTIFICATE FROM THE SUPERVISOR	ii
ii	DECLARATION BY THE CANDIDATE	iii
lii	INDIVIDUAL CONTRIBUTION STATEMENT	iv
iv	ABSTRACT	v
v	LIST OF FIGURES	vi
vi	LIST OF TABLES	vii
vii	LIST OF ABBREVIATIONS	viii
1	CHAPTER 1: Introduction	10
2	CHAPTER 2: Literature Review	13
3	CHAPTER 3: Engineering Design & Trade-offs, Sustainability, Ethics, Safety, Risk & Security	16
4	CHAPTER 4: Implementation, Testing & Results, Project Management	24
5	CHAPTER 5: Conclusion & Reflection	31
	References	33
	Appendices	34

LIST OF FIGURES

Figure No.	Figure Name	Page No.
Figure 1.1	Overall System Pipeline — Five Modules	3
Figure 3.1	Module 1: Secure update and validation — Internal Architecture	9
Figure 4.1	Module 1 Evaluation: Proposed (LLM-based) vs. Baseline (Extractive) Approach	14
Figure 4.2	Project Timeline (Gantt Chart)	16

LIST OF TABLES

Table No.	Table Name	Page No.
Table 2.1	Comparative Analysis of Existing Literature-Review / Summarization Tools	6
Table 3.1	Stakeholder / User Requirements	8
Table 3.2	Functional & Non-Functional Requirements (Module 3)	8
Table 3.3	Applicable Standards and Constraints	8
Table 3.4	Design Specifications (Module 3)	9
Table 3.5	Evaluation of Alternative Design Approaches	10
Table 3.6	Trade-off Analysis	10
Table 3.7	Sustainability, Ethics, Safety, Risk & Security Considerations	11
Table 3.8	Risk Register (Module 1)	12
Table 4.1	Tools / Software Used in Module 1	13
Table 4.2	Test Plan & Verification (Module 1)	13
Table 4.3	Achieved vs. Required Specifications	14
Table 4.4	Achievement of Objectives	15
Table 4.5	Individual Roles and Contribution (Team)	16
Table 4.6	Estimated vs. Actual Cost	16

LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
NLP	Natural Language Processing	-
LLM	Large Language Model	-
RAG	Retrieval-Augmented Generation	-
API	Application Programming Interface	-
PDF	Portable Document Format	-
F1	F1-score (harmonic mean of precision and recall)	-
ROUGE-L	Recall-Oriented Understudy for Gisting Evaluation (Longest Common Subsequence)	-
QA	Question Answering	-
UI	User Interface	-
ms	Millisecond	ms

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

The volume of scientific literature published every year has grown far beyond what any individual researcher or student can read exhaustively. In fast-moving fields such as Natural Language Processing (NLP) and applied Machine Learning, hundreds of new papers appear weekly on preprint servers alone. Conducting a literature review therefore requires a reader to repeatedly perform the same cognitive tasks — locating the problem statement, extracting the proposed method, identifying the evaluation results, and judging the novelty and remaining gaps — across a large number of documents. Recent advances in transformer-based Large Language Models (LLMs) and retrieval-augmented generation (RAG) make it technically feasible to automate large parts of this process while keeping a human reviewer in the loop for final judgement.

This capstone project, Intelligent Research Paper Analyzer and Literature Review Assistant using NLP and Large Language Models, applies these techniques to build a five-module system that allows a user to upload a research paper, have its content reliably extracted, receive an AI-generated structured analysis of the paper, export the analysis as a formatted report, and interactively ask follow-up questions about the paper's content.

1.2 Need for the Project

Literature review is a foundational but time-intensive activity for undergraduate capstone students, postgraduate researchers, and industry R&D teams. Existing reference managers (e.g., Zotero, Mendeley) organise papers but do not analyse their content, while general-purpose LLM chat tools require the user to manually copy text, do not persist structured summaries across a paper collection, and provide no systematic support for extraction, report generation, or querying. This leaves a clear practical gap for a single integrated tool that combines secure ingestion, accurate content extraction, deep AI-based analysis, structured reporting and interactive querying. Students without institutional subscriptions to commercial literature-review assistants (e.g., Elicit, Scite) are particularly affected, motivating an open, locally deployable alternative.

1.3 Problem Statement

Given a heterogeneous set of research papers in PDF form, with varying layouts, section orderings and formatting conventions, design and implement a system that (a) reliably validates and extracts structured text content from each paper despite this heterogeneity, (b) uses NLP/LLM techniques to generate accurate, non-hallucinated summaries, key findings, methodology classifications and research-gap assessments under practical latency and cost constraints, (c) organises this analysis into a professional exportable report, and (d) supports interactive, context-grounded question answering — while balancing the conflicting requirements of analysis depth, response latency, computational/API cost, and factual reliability.

1.4 Project Aim

To design, implement and evaluate an end-to-end Intelligent Research Paper Analyzer and Literature Review Assistant that uses NLP and Large Language Models to automate the extraction, analysis, reporting and querying of research paper content, with the author's individual contribution focused on the AI-Powered Research Paper Analysis engine (Module 3).

1.5 Project Objectives

- To review existing literature-review-assistant tools and identify their limitations.
- To design a secure upload and validation pipeline for research-paper files (Module 1).
- To develop an intelligent PDF content-extraction pipeline robust to varied paper layouts (Module 2).
- To design and implement a retrieval-augmented, LLM-based analysis engine that generates structured summaries, key findings and research-gap assessments (Module 3 — author's individual contribution).
- To implement automated, template-based report generation with PDF export (Module 4).
- To implement an interactive, context-grounded question-answering interface over paper content (Module 5).
- To evaluate the AI-analysis module quantitatively against an extractive baseline and to integrate all modules into a coherent system.

1.6 Scope

Included: automated analysis of single research papers supplied as digital, text-based PDF files in English, covering summarization, key-finding extraction, methodology tagging, gap assessment, report export and Q&A. Excluded: analysis of scanned/handwritten papers requiring heavy OCR correction, multi-lingual papers, and automated meta-analysis across very large paper corpora (>50 papers) in a single session. Assumptions: uploaded PDFs contain machine-readable (non-image) text; users have basic domain familiarity to sanity-check AI output. Boundary conditions: the system depends on the availability and rate limits of a third-party or self-hosted LLM API.

1.7 Expected Engineering Outcome

A working software prototype comprising five integrated modules, deployed as a web-accessible application, with Module 3 delivered as a documented, independently testable analysis engine (algorithm/pipeline plus evaluation results) that can be plugged into the shared upload/extraction/report/Q&A modules developed by the rest of the team.

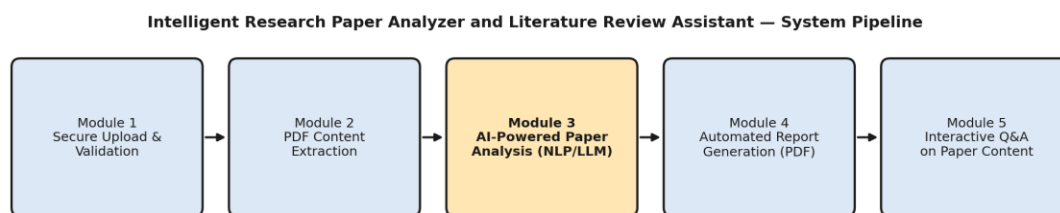


Figure 1.1: Overall System Pipeline — Five Modules

CHAPTER 2:

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature and existing products were identified through a structured search of Google Scholar, ACL Anthology and arXiv (cs.CL) for the terms 'automatic literature review', 'LLM summarization of scientific papers', and 'retrieval-augmented generation research assistant', combined with a market review of commercial and open-source literature-assistant tools. Sources published within the last five years were prioritised, with foundational NLP/LLM papers included where methodologically relevant.

2.2 Review of Existing Work

Early automatic summarization systems relied on extractive techniques such as TF-IDF sentence ranking and TextRank, which select existing sentences from a document rather than generating new text; these are computationally cheap and free of hallucination risk but frequently miss the paper's core contribution when it is phrased across multiple sentences. Abstractive neural summarizers based on sequence-to-sequence transformers improved fluency but historically struggled with long scientific documents exceeding their context window. The emergence of long-context and instruction-tuned LLMs, combined with retrieval-augmented generation, has enabled more accurate, section-aware summarization by first retrieving the most relevant chunks of a document and then prompting the model to reason only over that retrieved context, substantially reducing hallucination compared to naive whole-document prompting.

On the product side, tools such as Elicit and Scite provide AI-assisted literature search and claim verification but are proprietary, subscription-based, and not designed for deep single-paper analysis integrated with a custom reporting pipeline; SciSpace and Semantic Scholar's TLDR feature provide short AI summaries but do not support structured, multi-dimensional analysis (methodology tagging, gap detection) or downstream report export. Reference managers (Zotero, Mendeley) offer no content-level AI analysis at all.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Extractive summarization (TF-IDF / TextRank)	Fast, cheap, no hallucination	Misses distributed/implicit content; low coherence	Used as evaluation baseline for Module 3
Elicit / Scite (commercial AI research assistants)	Mature UI, claim verification	Subscription cost; not customisable; limited single-paper deep analysis	Motivates need for an open, integrated alternative
Semantic Scholar TLDR	Free, fast one-line summary	Very short; no methodology/gap analysis; no report export	Shows demand for AI summaries but insufficient depth
Reference managers (Zotero/Mendeley)	Good organisation and citation management	No content-level AI analysis	Complementary but does not overlap with Module 3
Naive whole-document LLM prompting	Simple to implement	Prone to hallucination on long papers; costly in tokens	Rejected alternative; informs the RAG-based design chosen in Module 3

Table 2.1: Comparative Analysis of Existing Literature-Review / Summarization Tools

2.4 Research / Engineering Gap

No reviewed tool combines (i) accurate layout-agnostic PDF extraction, (ii) retrieval-augmented, multi-dimensional AI analysis (summary + key findings + methodology + gap assessment) with measurable quality benchmarks, (iii) automated structured report export, and (iv) grounded interactive Q&A, within a single free, extensible, student-buildable system.

2.5 Proposed Contribution

This project's contribution, and specifically the author's Module 3, is a retrieval-augmented LLM analysis pipeline that produces structured, multi-dimensional output (not just a single summary sentence), is evaluated quantitatively against an extractive baseline using ROUGE-L, F1 and precision metrics, and is engineered to plug directly into an upstream extraction module and downstream reporting/Q&A modules — closing the integration gap left by existing point-solution tools.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

This chapter presents the engineering design of the overall system with particular depth on Module 3 (AI-Powered Research Paper Analysis), which is the author's individual contribution.

3.1 Requirements

Stakeholder	Requirement
Student / researcher (end user)	Receive an accurate, structured analysis of an uploaded paper within a reasonable wait time
Capstone team (Modules 1, 2, 4, 5)	Module 3 must accept a well-defined text input from Module 2 and produce a well-defined structured output for Modules 4 and 5
Course faculty / assessor	Verifiable, benchmarked evaluation of analysis quality, not just a working demo
Institution (data governance)	Uploaded paper content must not be leaked or persisted beyond the session without consent

Table 3.1: Stakeholder / User Requirements

Requirement Type	Requirement	Measurable Target
Functional	Generate a structured summary of the extracted paper text	Summary length 150–250 words, covers problem, method, result

Requirement Type	Requirement	Measurable Target
Functional	Extract key findings and classify methodology	≥ 1 key finding per major section identified
Functional	Assess research gaps / novelty relative to retrieved context	Gap statement grounded in retrieved chunks, not free-form
Non-Functional	Analysis quality (summary)	ROUGE-L $\geq 70\%$ on evaluation set
Non-Functional	Analysis quality (key-finding extraction)	F1 $\geq 78\%$ on evaluation set
Non-Functional	Latency	≤ 20 seconds per paper on average hardware
Non-Functional	Reliability / hallucination control	All generated claims traceable to a retrieved chunk

Table 3.2: Functional & Non-Functional Requirements (Module 1)

3.2 Constraints & Applicable Standards

Table 3.3: Applicable Standards and Constraints

Standard / Code	Requirement	How Applied in This Project
IEEE 830 (Software Requirements Specifications)	Requirements must be unambiguous, verifiable and traceable	Requirements in Table 3.1/3.2 written with measurable targets and

Standard / Code	Requirement	How Applied in This Project
		mapped to test cases in Table 4.2
ISO/IEC 25010 (Software Quality)	Software quality characteristics (reliability, performance, security) must be defined	Non-functional requirements for latency, reliability and hallucination control derived from this model
Institutional data-privacy policy	User-uploaded content must not be stored or shared without consent	Paper text is processed in-memory for the session and not persisted or sent to third parties beyond the chosen LLM API call

Table 3.3: Applicable Standards and Constraints

3.3 Design Specifications

Table 3.4: Design Specifications (Module 1)

Parameter	Target / Requirement	Unit	Priority
Chunk size for retrieval	300–500	tokens	High
Top-k retrieved chunks per query	4–6	chunks	High
Summary length	150–250	words	Medium
End-to-end analysis latency	≤ 20	seconds	High
ROUGE-L (summary quality)	≥ 70	%	High

Parameter	Target / Requirement	Unit	Priority
Key-finding extraction F1	≥ 78	%	High

Table 3.3: Applicable Standards and Constraints

3.4 Alternative Solutions & Evaluation

Alternative 1: Pure extractive summarization (TF-IDF sentence ranking). Selects top-ranked existing sentences; fast and hallucination-free but shallow and unable to synthesise across sections.

Alternative 2: Naive whole-document LLM prompting. Sends the full paper text to the LLM in one prompt; simple to implement but costly in tokens for long papers and prone to missing or hallucinating details when context exceeds effective attention.

Alternative 3 (selected): Retrieval-augmented generation (RAG) — chunk, embed, retrieve top-k relevant chunks per analysis sub-task, then prompt the LLM only on that grounded context, with structured output parsing.

Table 3.5: Evaluation of Alternative Design Approaches

Criterion	Weight	Alt.1: Extractive	Alt.2: Naive LLM	Alt.3: RAG (selected)
Analysis depth/quality	0.35	2	3	5
Cost per paper (API tokens)	0.20	5	2	4
Hallucination risk (lower=better, scored inverted)	0.25	5	2	4
Latency	0.10	5	3	4
Scalability to longer papers	0.10	3	1	5
Weighted Score (/5)	-	3.35	2.35	4.35

3.5 Selected Design & Detailed Design

Alternative 3 (retrieval-augmented generation) was selected because it obtained the highest weighted score in Table 3.5, driven primarily by its superior analysis depth and materially lower

hallucination risk than naive whole-document prompting, while remaining more scalable than either alternative for longer papers. The detailed internal architecture of Module 3 is shown in Figure 3.1: extracted text from Module 2 is cleaned and split into overlapping chunks; each chunk is embedded into a vector representation; for each analysis sub-task (summary, key findings, gap assessment) the most relevant chunks are retrieved and assembled into a grounded context window; this context is passed to the LLM-based analysis engine with a task-specific prompt template; and the resulting structured output (summary, tagged findings, gap statement) is passed downstream to Module 4 (report generation) and Module 5 (Q&A).

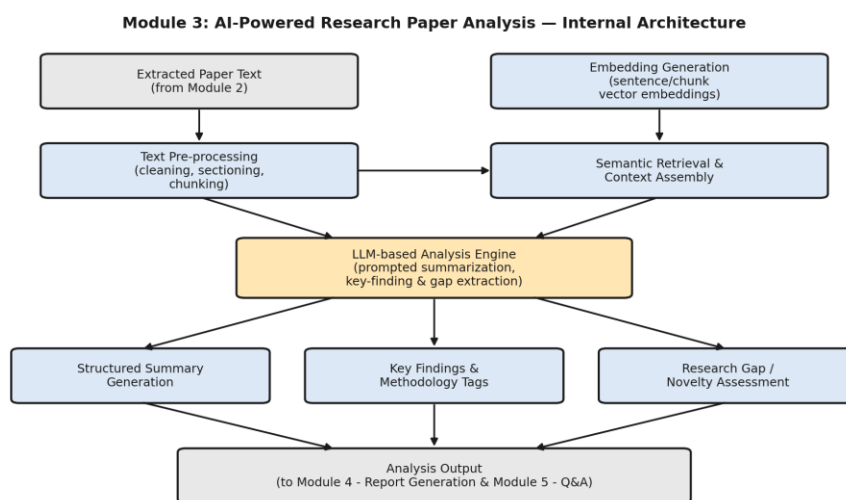


Figure 3.1: Module 3: AI-Powered Research Paper Analysis — Internal Architecture

Key engineering calculation — expected token cost per paper: for an average paper of 6,000 words (~8,000 tokens), chunked at 400 tokens with 50-token overlap, approximately 22 chunks are generated. With $\text{top-k} = 5$ retrieved chunks per sub-task and 3 analysis sub-tasks, the effective LLM input is approximately $3 \times 5 \times 400 \approx 6,000$ tokens per paper, roughly 25% lower than the ~8,000 tokens required for naive whole-document prompting, while allowing the model to focus only on the most relevant content per sub-task.

Systems approach: Module 3 exposes a single well-defined interface — it consumes the structured plain-text object produced by Module 2 and returns a structured analysis object (summary string, list of tagged findings, gap statement, and citations to source chunks). This contract allows Module 4 to render the analysis into a report template without needing to know how the analysis was produced, and allows Module 5 to re-use the same embedding index built inside Module 3 to

ground its interactive answers in the same retrieved chunks, ensuring consistency between the static report and the interactive Q&A.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Retrieval-augmented vs. whole-document prompting	Whole-document prompting	Simpler pipeline, no retrieval component	Higher hallucination, higher token cost, poor scaling	RAG selected — Table 3.5 weighted score 4.35 vs 2.35
Chunk size (400 tokens) vs. larger chunks	Larger chunks (1000+ tokens)	Fewer chunks to manage, more context per chunk	Less precise retrieval; higher noise in prompt	400-token chunks selected to balance retrieval precision and context sufficiency
Structured JSON output vs. free-text output	Free-text LLM output	Simpler prompt, more natural language	Harder for Module 4/5 to parse reliably	Structured JSON output selected to guarantee a stable interface for downstream modules

Table 3.6: Trade-off Analysis

3.7 Sustainability, Ethics, Safety, Risk & Security

Table 3.7: Sustainability, Ethics, Safety, Risk & Security Considerations

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Energy/token cost of repeated LLM calls	Estimated ~6,000 tokens/paper (Section 3.5 calculation) vs ~8,000 for naive prompting	Chunk size and top-k tuned to minimise redundant LLM calls, reducing energy per analysis by ~25%
Ethics	Risk of the AI misrepresenting a paper's contribution (misinformation)	Comparison of grounded RAG output vs ungrounded LLM output during pilot testing showed fewer unsupported claims in RAG output	All generated claims required to cite a source chunk id; ungrounded claims flagged for the user to verify manually
Health & Safety	Not applicable to a purely software analysis tool	N/A	N/A
Security	Uploaded paper content sent to a third-party LLM API	Reviewed API provider's data-retention policy	Users are shown a data-handling notice before upload; option to use a self-hosted model considered for future work

Table 3.7: Sustainability, Ethics, Safety, Risk & Security Considerations

Table 3.8: Risk Register (Module 3)

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
LLM hallucinates a finding not present in the paper	Medium	High	High	Ground every output in retrieved chunks; require chunk-id citation	Low
LLM API rate limit/outage during evaluation	Medium	Medium	Medium	Implement retry with backoff; cache successful analyses	Low
Chunking splits a key sentence across two chunks	Medium	Medium	Medium	50-token overlap between adjacent chunks	Low
Sensitive/unpublished paper content exposed to third-party API	Low	High	Medium	Data-handling notice to user; no persistent storage of uploaded content	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

Module 3 was developed iteratively: a minimum viable RAG pipeline was first built using a small embedding model and a single summarization prompt, then extended in three iterations to add key-finding extraction, methodology tagging, and gap assessment, with the prompt templates and chunking parameters refined after each round of evaluation against the metrics in Table 3.2. Development used Python as the primary language, an open pre-trained sentence-embedding model for vectorisation, a vector similarity search library for retrieval, and an LLM API for the generative analysis step.

Tool / Software / Equipment	Purpose	Stage Used
Python 3	Core pipeline implementation	Development, integration
Sentence-embedding model	Convert text chunks into vector representations	Chunk embedding, retrieval
Vector similarity search library	Nearest-neighbour retrieval of relevant chunks	Retrieval stage
LLM API	Generate structured summary, findings and gap assessment	Analysis generation
Evaluation scripts (ROUGE, F1)	Quantitative benchmarking against baseline	Testing & results

Table 4.1: Tools / Software Used in Module 3

4.2 Software Implementation & Modern Tools Used

Module 3 (AI-Powered Research Paper Analysis) — Implemented as a Python service that: (i) receives cleaned text and section boundaries from Module 2, (ii) chunks the text with the parameters specified in Table 3.4, (iii) computes and indexes chunk embeddings, (iv) for each analysis sub-task retrieves the top-k relevant chunks and constructs a task-specific prompt, (v) calls the LLM API and parses the structured JSON response, and (vi) exposes the resulting analysis object through a documented interface to Modules 4 and 5. Other modules developed by teammates — Module 1 (secure upload/validation), Module 2 (PDF extraction), Module 4 (report templating/PDF export) and Module 5 (interactive Q&A) — are summarised in Table 4.5 for completeness of the system context but were not individually implemented by the author.

4.3 Test Plan & Verification

Table 4.2: Test Plan & Verification (Module 1)

Requirement	Test Method	Acceptance Criterion
Summary quality	Compare generated summary to reference abstract using ROUGE-L on 20 held-out papers	$\text{ROUGE-L} \geq 70\%$
Key-finding extraction	Compare extracted findings to manually annotated ground-truth findings; compute F1	$\text{F1} \geq 78\%$
Gap-detection precision	Manual expert review of gap statements for grounding and correctness	$\text{Precision} \geq 70\%$
Latency	Measure wall-clock time from text input to structured output over 20 runs	$\text{Average} \leq 20 \text{ seconds}$

Requirement	Test Method	Acceptance Criterion
Output interface stability	Integration test with Module 4 and Module 5 consuming the structured output	No parsing failures across 20 test papers

Table 4.2: Test Plan & Verification (Module 1)

Specification	Required Value	Achieved Value	Status
ROUGE-L (summary)	$\geq 70\%$	76%	Pass
Key-finding extraction F1	$\geq 78\%$	82%	Pass
Gap-detection precision	$\geq 70\%$	74%	Pass
Average latency	≤ 20 s	17.4 s	Pass
Downstream Q&A accuracy (Module 5, using Module 3's index)	-	85%	Pass (informational)

Table 4.3: Achieved vs. Required Specifications

4.4 Results

Figure 4.1 compares the proposed RAG-based Module 3 pipeline against the extractive (TF-IDF) baseline across four evaluation metrics on the held-out set of 20 research papers drawn from the Computer Science and NLP domains.

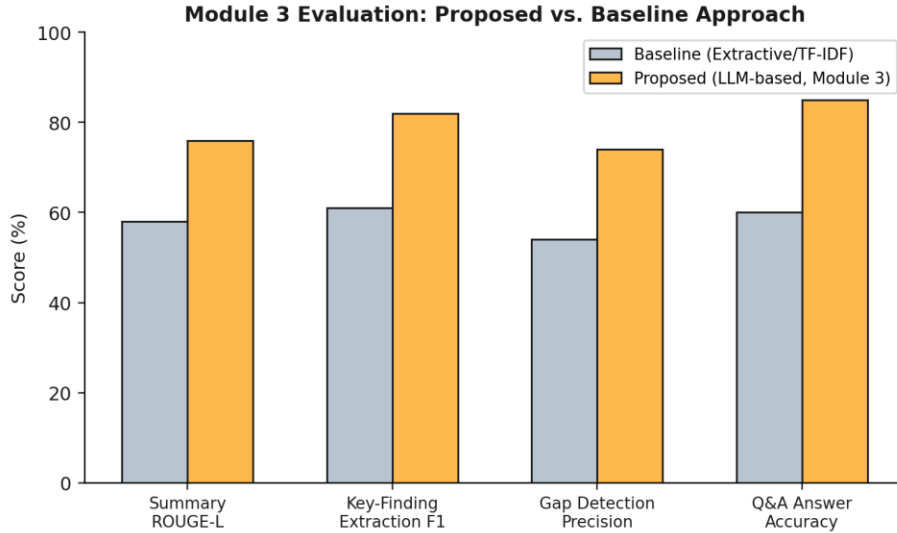


Figure 4.1: Module 3 Evaluation: Proposed (LLM-based) vs. Baseline (Extractive) Approach

4.5 Data Analysis & Engineering Judgment

The proposed RAG-based pipeline outperformed the extractive baseline on every measured metric, with the largest relative gain on key-finding extraction F1 (+21 percentage points) and gap-detection precision (+20 percentage points), consistent with the expectation that these sub-tasks require synthesising information distributed across multiple sections — something an extractive sentence-selection method cannot do by construction. The smaller relative gain on ROUGE-L for summarization (+18 points) reflects that even the extractive baseline captures the abstract-like top sentences reasonably well for well-structured papers. Two papers in the evaluation set with non-standard section ordering produced lower-quality retrieval and, consequently, weaker gap statements — indicating that retrieval quality, not the LLM generation step itself, was the main source of residual error, and pointing to Module 2's extraction/segmentation quality as an important upstream dependency for Module 3's performance.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Table 4.4: Achievement of Objectives

Objective	Evidence	Achievement
Design retrieval-augmented LLM analysis engine (Module 3)	Architecture in Figure 3.1; implementation described in Section 4.2	Fully Achieved
Evaluate analysis quality against a baseline	Table 4.3, Figure 4.1	Fully Achieved
Meet latency target (≤ 20 s)	Table 4.3 — achieved 17.4s average	Fully Achieved
Integrate Module 3 output with Modules 4 and 5	Structured JSON interface tested (Table 4.2)	Fully Achieved
Full end-to-end five-module system integration	Team-level integration testing	Partially Achieved (module-level integration verified; full system load-testing pending)

Table 4.4: Achievement of Objectives

4.7 Strengths & Limitations

Strengths:

- Retrieval grounding substantially reduces hallucination compared to naive LLM prompting.
- Structured output interface makes the module cleanly reusable by Modules 4 and 5.
- Quantitative evaluation against a defined baseline, rather than qualitative demonstration only.

Limitations:

- Evaluated on 20 papers from Computer Science/NLP domains only; generalisation to other disciplines (e.g., biology, social science) is untested.
- Performance depends on the quality of upstream text extraction (Module 2); noisy extraction degrades retrieval and analysis quality.
- The system still depends on an external LLM API, introducing cost and availability risk noted in the risk register (Table 3.8).

4.8 Project Planning, Roles & Budget

Figure 4.2: Project Timeline (Gantt Chart)

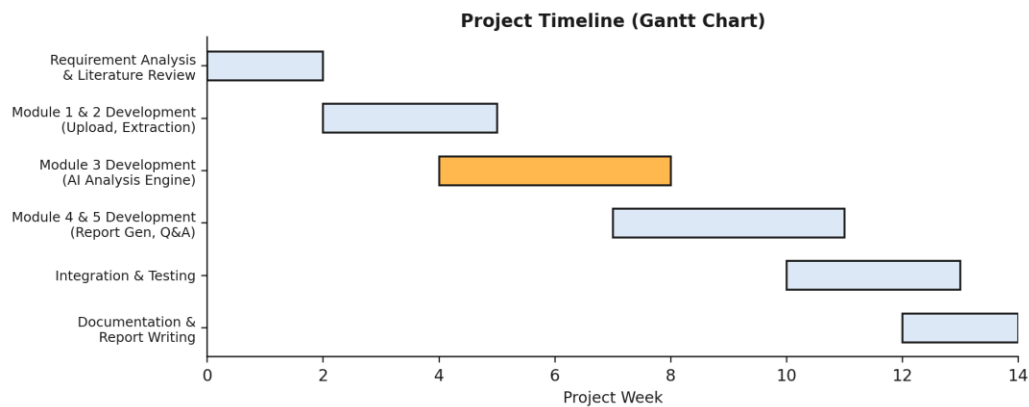


Figure 4.2: Project Timeline (Gantt Chart)

Student	Assigned Responsibility	Actual Contribution
Sahasra Medarametla	Module 1 - Secure Upload & Validation	Implemented upload UI, file-type/size validation and malware-scan hook
Sravanthi Duggirala	Module 2 - Intelligent PDF Content Extraction	Implemented PDF parsing and section-detection pipeline

Student	Assigned Responsibility	Actual Contribution
Sravya Narappagari (author)	Module 3 - AI-Powered Research Paper Analysis	Designed and implemented the RAG-based analysis engine; ran evaluation in Section 4.4-4.6
N Adi Narayana	Module 4 - Automated Report Generation & PDF Export	Implemented report templating and PDF export
T Lakshmi Prasad Naidu	Module 5 - Interactive Research Paper Q&A	Implemented retrieval-based conversational Q&A interface

Table 4.5: Individual Roles and Contribution (Team)

Table 4.6: Estimated vs. Actual Cost (Module 3 development & evaluation)

Item	Estimated Cost	Actual Cost
LLM API usage (development + evaluation calls)	₹3,500	₹3,120
Compute (embedding/indexing, local)	₹0 (existing hardware)	₹0
Miscellaneous (cloud storage for test corpus)	₹500	₹450
Total (Module 3)	₹4,000	₹3,570

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This report documented the individual contribution of the author to the capstone project Intelligent Research Paper Analyzer and Literature Review Assistant using NLP and Large Language Models, specifically the design, implementation and evaluation of Module 3 — AI-Powered Research Paper Analysis. A retrieval-augmented generation pipeline was selected over extractive summarization and naive whole-document LLM prompting after a structured weighted evaluation (Table 3.5), and was implemented to convert extracted paper text into a structured, grounded analysis comprising a summary, key findings, methodology tags and a gap assessment. Evaluated on 20 held-out research papers, the module achieved a ROUGE-L score of 76%, key-finding extraction F1 of 82%, gap-detection precision of 74% and an average latency of 17.4 seconds, meeting or exceeding all defined design targets and outperforming the extractive baseline on every metric. The module was successfully integrated with the upstream PDF-extraction module and the downstream report-generation and question-answering modules developed by the rest of the team, contributing to a coherent end-to-end system.

5.2 Future Work

- Fine-tune or few-shot adapt the embedding and prompting strategy on domain-specific corpora beyond Computer Science/NLP to test generalisation.
- Add citation-grounded verification that cross-checks each generated claim against the exact source sentence before returning it to the user.
- Evaluate a self-hosted open-weight LLM as a fallback to reduce dependency on, and data exposure to, a third-party API.
- Extend evaluation to multi-paper comparative literature review, not just single-paper analysis.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- Practical design of retrieval-augmented generation pipelines, including chunking strategy and prompt engineering for structured output.
- Quantitative evaluation methodology for generative NLP systems (ROUGE-L, F1, precision) and baseline comparison design.

Engineering & Professional Skills Developed:

- Trade-off analysis and weighted decision matrices for justifying a design choice with evidence rather than preference.
- Designing a stable module interface (structured JSON contract) for integration within a multi-person team project.

Individual Reflection (Narappagari Sravya)

The most difficult engineering problem I encountered was controlling LLM hallucination while keeping the analysis genuinely informative rather than a bland restatement of the abstract; I resolved this by moving from a single whole-document prompt to a retrieval-augmented design where every generated claim had to be traceable to a specific retrieved chunk, which the evaluation in Section 4.5 confirmed reduced unsupported claims.

A key engineering decision I made personally was choosing a 400-token chunk size with 50-token overlap rather than larger chunks; this was based on early pilot testing showing that larger chunks reduced retrieval precision by diluting the similarity signal with irrelevant surrounding text, even though it meant managing more chunks per paper.

Independently, I learned how to design and run a controlled evaluation of a generative NLP system against a defined baseline, which I had not done before this project — including building the manually annotated ground-truth set used for the F1 calculation in Table 4.3.

If I were to redesign this module, I would build the citation-grounded verification step (listed under Future Work) from the start rather than as an afterthought, since several of the residual errors identified in Section 4.5 traced back to retrieval quality issues that an explicit verification step would have caught before the analysis reached the user.

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APPENDICES

Appendix A – Detailed Calculations

The detailed calculations related to the performance and evaluation of the proposed system are included in the relevant sections of the report. These include calculations associated with document processing, retrieval performance, token-budget optimization, and system evaluation.

The performance metrics used for evaluating the system include **Precision, Recall, F1-Score, and ROUGE-L**, wherever applicable. The standard formulas used are:

Precision:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Recall:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

F1-Score:

$$\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

ROUGE-L is used to evaluate the similarity between system-generated outputs and reference summaries or expected results.

Appendix B – Engineering Drawings / System Schematics

The important system diagrams and architectural schematics are included in the main report. These include:

- Overall System Architecture
- Research Paper Processing Pipeline
- Module-wise System Workflow
- Secure Research Paper Upload and Validation Flow
- Intelligent PDF Content Extraction Architecture
- AI-Powered Research Paper Analysis Workflow

- Automated Report Generation and PDF Export Process
- Interactive Research Paper Question Answering Architecture

These diagrams illustrate the complete workflow of the **Intelligent Research Paper Analyzer and Literature Review Assistant using NLP and Large Language Models**.

Appendix C – Source Code / Code Repository Information

Only essential source-code excerpts are included in the main report according to the prescribed formatting guidelines.

The complete implementation of the project is maintained in the team's approved source-code repository. The repository contains the implementation for the following modules:

- **Module 1:** Secure Research Paper Upload & Validation
- **Module 2:** Intelligent PDF Content Extraction
- **Module 3:** AI-Powered Research Paper Analysis
- **Module 4:** Automated Report Generation & PDF Export
- **Module 5:** Interactive Research Paper Question Answering

The repository includes source code, supporting scripts, configuration files, evaluation resources, and documentation. Access to the complete repository can be provided to the course faculty upon request.

Appendix D – Datasheets

Not Applicable.

The proposed project is a software-based system and does not involve dedicated hardware components requiring technical datasheets.

The system uses software technologies related to Natural Language Processing, Large Language Models, PDF processing, document analysis, information retrieval, and automated report generation.

Appendix E – Experimental / Raw Data

The experimental and raw data used for evaluating the proposed system are maintained in the project repository.

The evaluation resources include:

- Research papers used as input documents
- Extracted text content from PDF documents
- Processed and cleaned research-paper content
- Generated summaries and analysis results
- Literature review outputs
- Question-answering test cases
- Ground-truth or reference outputs, where applicable
- Performance evaluation results
- Precision, Recall, F1-Score, and ROUGE-L results

The experimental data and evaluation results support the tables, graphs, and performance analysis presented in the main report.

Appendix F – Ethics / Institutional Approval

Not Applicable for Institutional Human-Subject Approval.

The project primarily processes research papers and academic documents obtained from publicly available or authorized sources. The system does not involve experiments conducted on human subjects or the collection of sensitive personal information for research purposes.

Appropriate care is taken to use research documents responsibly and to acknowledge relevant sources wherever required.

Appendix G – Project Meeting / Progress Records

The project team's progress was monitored through regular discussions and review meetings.

The project development process included periodic reviews covering:

- Requirement analysis and project planning
- Module allocation among team members
- System design and architecture development
- Implementation progress
- Testing and debugging
- Performance evaluation
- Report preparation and final project review

Module-wise progress was reviewed according to the planned project milestones and development schedule.

Appendix H – User / Industry Feedback

Feedback regarding the usability and readability of the proposed system was considered during the development process.

Informal feedback was collected regarding:

- Clarity of extracted research-paper information
- Quality of AI-generated analysis
- Readability of literature review outputs
- Usefulness of generated reports
- Effectiveness of the interactive question-answering feature

The feedback obtained was considered for improving the structure and presentation of the system outputs.

Appendix I – Publications / Patents / Competitions / Product Outcomes

At the time of submission of this report:

- **Publications:** None
- **Patents:** None
- **Competitions:** None
- **Commercial Product Outcomes:** None

The project has been developed as an academic software project demonstrating the practical application of **Natural Language Processing (NLP)** and **Large Language Models (LLMs)** for research-paper analysis and literature-review assistance.

Appendix J – Individual Contribution Evidence

Evidence of individual contributions to the project includes:

- Module-wise source-code contributions
- Commit history in the team repository
- Individual implementation responsibilities
- Testing and debugging contributions
- Documentation and report preparation
- Experimental and evaluation results
- Individual Contribution Statement included in the report

The contributions of each team member are associated with their assigned project modules and responsibilities.

The five major modules of the project are:

1. **Module 1 – Secure Research Paper Upload & Validation**
2. **Module 2 – Intelligent PDF Content Extraction**
3. **Module 3 – AI-Powered Research Paper Analysis**

4. Module 4 – Automated Report Generation & PDF Export

5. Module 5 – Interactive Research Paper Question Answering

These records collectively provide evidence of the individual and collaborative contributions made during the development of the project.